

Team Name:

Team #:

Hawk-Hornet Invitationals

November 15, 2025

Materials Science Division C

Team Number: _____

School: _____

Student: _____

Student: _____

Written Test Score Total Points: 96__

Written Test Score Total Tie Breaker Points: __6__

Instructions:

- *Each participant must bring a writing implement and a stand-alone calculator of any type (Class III).*
- *Each participant may bring one unique 8.5" x 11" sheet of paper with information on both sides in any form and from any source.*
- *There will be 25 minutes to take the test.*

GOOD LUCK! 😊

1. The word 'Nano' comes from a Greek root that means what? (2 pts)
2. A new multifunctional wearable sensor is being designed. Based on mechanical flexibility, electrical conductivity, and biocompatibility, which material would not work well? (2 pts)
 - a. Graphene
 - b. Carbon nanotubes
 - c. Fullerenes
 - d. Carbon Nanofibers
 - e. None of the above
3. Which nanomaterial is most likely used in solar cells? (2 pts)
 - a. Metal nanomaterials
 - b. Metal-oxide nanomaterials
 - c. Ceramic nanomaterials
 - d. Semiconductor nanomaterials
 - e. Carbon-based nanomaterials
 - f. None of the above
4. Which of the following nanomaterials can't be used in drug delivery? (2 pts)
 - a. Metal nanomaterials
 - b. Metal-oxide nanomaterials
 - c. Ceramic nanomaterials
 - d. Semiconductor nanomaterials
 - e. Carbon-based nanomaterials
 - f. None of the above
5. What is doping? (2 pts)
6. List three applications of semiconductors. (6 pts)

7. What is an intrinsic semiconductor? (2 pts)
8. How do you distinguish the dimensionality of a nanomaterial? (2 pts)
9. What is the energy band gap? (2 pts)
10. What characterization determines the energy band gap? (2 pts)
11. Categorize the following: metal (MN), Metal-oxide (MO), Ceramic (C), Semiconductor (S), and Carbon-based (CB) nanomaterials (15 pts)
 - a. Titanium dioxide nanoparticles _____
 - b. Graphene nanoparticles _____
 - c. Copper nanoparticles _____
 - d. Cadmium selenide quantum dots _____
 - e. Zirconia nanoparticles _____
 - f. Silver nanoparticles _____
 - g. Diamond nanoparticles _____
 - h. Zinc sulfide nanoparticles _____
 - i. Graphene oxide _____
 - j. Silicon carbide nanoparticles _____
 - k. Iron oxide nanoparticles _____
 - l. Gold nanoparticles _____
 - m. Silicon nanoparticles _____
 - n. Alumina nanoparticles _____
 - o. Zinc oxide nanoparticles _____
12. What is a common application of nanofluids? (2 pts)
 - a. Rocket fuel
 - b. Heat exchanger coolant
 - c. Building material
 - d. Data storage in hard drives
 - e. All the above

13. What property of nanomaterials is most likely to change as particle size decreases below 100 nm? (2 pts)

- a. Density
- b. Electrical resistivity
- c. Elasticity
- d. Strength
- e. All of the above

14. Which type of force is primarily responsible for holding atoms together in ceramic nanomaterials? (2 pts)

- a. Hydrogen bonding
- b. Van der Waals forces
- c. Covalent bonding
- d. Metallic bonding

15. Which of the following is/are intermolecular forces? (2 pts)

- a. Hydrogen bonding
- b. Van der Waals forces
- c. Covalent bonding
- d. Metallic bonding
- e. Ionic bonding

16. Which of the following is/are true? (2 pts)

- a. Grain boundaries decrease the surface area of nanomaterials.
- b. Grain boundaries increase strength as barriers to dislocation movement
- c. Grain boundaries always reduce electrical conductivity in nanomaterials.
- d. Grain boundaries eliminate quantum size effects in nanomaterials.

17. A droplet of water is placed on a nanomaterial surface. The surface tension of water is 72.8 mN/m, the surface energy of the solid is 40 mN/m, and the interfacial tension between the water and solid is 30 mN/m. (6 pts)

17a. Use Young's equation to calculate the wetting angle: $\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos(\theta)$

17b. Is the surface hydrophobic or hydrophilic? How do you know?

18. What is the primary role of surfactants in a liquid? (2 pts)

- a. Increase the liquid's density
- b. Decrease the liquid's surface tension
- c. Increase the liquid's boiling point
- d. Increase the liquid's viscosity

19. Why is it called top-down and bottom-up synthesis? (2 pts)

20. Which of the following is/are true of a sol-gel process? (2 pts)

- a. A method where a colloidal solution (sol) gradually forms a gel-like network, used to produce ceramics, glasses, and nanostructured materials.
- b. A high-temperature melting technique used to produce single-crystal nanomaterials.
- c. A bottom-up chemical process where metal alkoxides or salts undergo hydrolysis and condensation to form nanostructured oxides.
- d. A physical vapor deposition method where thin films are formed by evaporation under vacuum

21. If you needed to fabricate high-precision nanoscale circuits for a computer chip, which technique would you most likely use? (2 pts)

- a. Lithography
- b. Sol-gel process
- c. Mechanical milling
- d. Molecular self-assembly

22. You need to know the crystal structure of an unknown material and the elemental composition. Which two characterization techniques would you need to use? (4 pts, tiebreaker)

23. What type(s) of spectroscopy could you use to find chemical bonds and functional groups? (3 pts)

- a. UV-Visible Spectroscopy
- b. Photoluminescence Spectroscopy
- c. Energy Dispersive X-ray Spectroscopy
- d. Mass Spectroscopy
- e. Fourier Transform Infrared Spectroscopy
- f. Raman Spectroscopy
- g. X-ray Photoelectron Spectroscopy

24. A sample of a crystalline material is analyzed using X-ray diffraction with X-ray wavelength of 0.200 nm. A diffraction peak is observed at a θ angle of 45° corresponding to a particular set of lattice planes. Calculate the interplanar spacing for these planes using Bragg's Law, including units. Bragg's Law: $n\lambda = 2d\sin(\theta)$. (2 pts)

25. What is the difference between adhesion and cohesion? (2 pts)

26. A nanocomposite coating is applied to reduce friction on a machine part. What causes this wear resistance? (2 pts, tiebreaker)

- a. Low density and low elastic modulus

- b. High hardness and strong interfacial bonding
- c. Low elastic modulus and high hardness
- d. High melting temperature and low density

27. A study found that a 1 micrometer long carbon nanotube with an outer diameter of 6 and inner diameter of 3nm deformed 0.01% when 3 nN was applied to the face of the cylinder. What is the elastic modulus, including units? (7 pts)

28. Light hits the chlorophyll of a leaf. What happens to the different wavelengths of light that causes a green appearance? (2 pts)

29. A beam of light with wavelength 400 nm and a refractive index $n=1.5$ passes through a colloidal suspension of nanoparticles of radius 50 nm. The scattering cross-section for Rayleigh scattering is:

$$\sigma = \frac{8\pi^3 r^6}{3\lambda^4} \left(\frac{(n^2 - 1)}{(n^2 + 2)} \right)^2$$

- a. Calculate the scattering cross-section σ of a single particle. (2 pts)

Team Name:

Team #:

b. Using the Beer–Lambert law for scattering, if the concentration N is $10^{16}m^{-3}$ and Length is 3 cm, what fraction of light is transmitted? Law: $I = I_0e^{-N\sigma L}$ (2 pts)

c. What is the fraction of light transmitted if the concentration is $10^{18}m^{-3}$? (2 pts)

True False:

1. Smaller nanomaterials have a higher surface area/volume ratio (1 pts)

True
False

2. Molecular beam epitaxy is a method of Bottom-up synthesis (1 pts)

True
False

3. The effective density of nanomaterials increases when porosity increases (1 pts)

True
False

4. Smaller powders will have a lower melting temperature(1 pts)

True
False

Team Name:

Team #:

5. Metals with high electrical conductivity will always have proportionally high thermal conductivity (1 pts)

True

False